

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on Python and SQL experience analyzing large, regulated datasets at Deutsche Bundesbank, plus independent projects spanning predictive modeling and data pipeline design. Comfortable turning raw, messy data into clear, decision-ready analysis, and detail-oriented by habit from building auditable data quality checks. Drawn to the energy transition as a domain where rigorous analysis has direct, measurable impact.

Core Competencies

- **Programming & Data Analysis:** Python (Pandas, NumPy, Scikit-Learn), SQL, R, statistical analysis of large datasets
- **Data Quality & Auditing:** Automated testing and error detection on regulated banking data (Deutsche Bundesbank AnaCredit); building traceable, auditable analysis workflows
- **Market & Trend Research:** Independent research combining quantitative analysis with clear written summaries for non-technical stakeholders
- **Reporting & Presentation:** Excel/Google Sheets, Tableau, Matplotlib, PowerPoint/Google Slides for translating analysis into decision-ready summaries
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, K-Means, predictive modeling
- **Languages & Collaboration:** Fluent English (C1, TOEFL iBT 105/120) and German (C1), Spanish (native); Git, Jira

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Built automated Python/Pandas testing (row counts, referential integrity, aggregate consistency) to validate business logic across large-scale reported banking data
 - Designed automated error detection for value-range violations and duplicate entries, with automated ticketing and stakeholder notifications, cutting response time to data quality issues
 - Worked directly with data supplied by external reporting banks, requiring careful validation before any conclusion could be trusted, a discipline directly transferable to market and load data analysis

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Designed an end-to-end pipeline for simulated bank transactions (150 accounts): normalized SQLite database, staged data cleaning, and a fully auditable change log
 - Produced quality visualizations and a structured exception report from the cleaned data
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data to identify trends and resource bottlenecks, using Random Forest, Neural Networks, and K-Means clustering

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- English (C1, TOEFL iBT 105/120), German (C1), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.